



Physics Assignment
Electromagnetic Induction

Section A - Multiple Choice Questions

Choose the correct option. Each question carries 1 mark.

- Q1. A bar magnet is dropped through a long vertical copper tube. The final acceleration of the magnet is: [1 Mark]
- A. equal to g B. less than g
C. greater than g D. zero
- Q2. A coil is placed in a region of magnetic field. The magnitude of induced EMF in the coil does not depend on: [1 Mark]
- A. number of turns in the coil
B. change in magnetic flux
C. time rate of change of magnetic flux
D. resistance of the coil
- Q3. A conducting rod of length ' l ' moves with velocity ' v ' perpendicular to a uniform magnetic field ' B '. The magnitude of the induced EMF is: [1 Mark]
- A. Blv B. $Blv \sin(\theta)$
C. B^2lv D. Bl/v
- Q4. The self-inductance of a coil is maximum when: [1 Mark]
- A. it has a large number of turns
B. it has a small number of turns
C. it is made of a non-magnetic material
D. it is very thin
- Q5. Two coils are wound on a non-conducting cylindrical core. If the current in the primary coil is suddenly switched off, the induced EMF in the secondary coil is: [1 Mark]
- A. zero B. maximum
C. minimum D. constant
- Q6. An inductor of 20 mH carries a current of 1 A. The energy stored in the inductor is: [1 Mark]
- A. 10 mJ B. 20 mJ
C. 10 J D. 20 J

- Q7. Which of the following is NOT an application of eddy currents? [1 Mark]
- A. Induction furnace
 - B. Magnetic braking in trains
 - C. Speedometer
 - D. Electric heater
- Q8. Assertion (A): Lenz's law is a consequence of the law of conservation of energy. Reason (R): The induced current always opposes the change in magnetic flux responsible for its production. [1 Mark]
- A. Both A and R are true and R is the correct explanation of A
 - B. Both A and R are true but R is not the correct explanation of A
 - C. A is true but R is false
 - D. A is false but R is true
- Q9. Assertion (A): Self-inductance is called the inertia of electricity. Reason (R): An inductor opposes any change in the current flowing through it. [1 Mark]
- A. Both A and R are true and R is the correct explanation of A
 - B. Both A and R are true but R is not the correct explanation of A
 - C. A is true but R is false
 - D. A is false but R is true
- Q10. A square loop of side 'a' is moving with a constant velocity 'v' in a uniform magnetic field 'B' perpendicular to the plane of the loop. The field exists only for a region of width '2a'. When the loop is just entering the magnetic field, the magnitude of the induced EMF is: [1 Mark]
- A. Bva
 - B. $2Bva$
 - C. Bv/a
 - D. 0
- Q11. A square loop of side 'a' is moving with a constant velocity 'v' in a uniform magnetic field 'B' perpendicular to the plane of the loop. The field exists only for a region of width '2a'. When the loop is completely inside the magnetic field, the magnitude of the induced EMF is: [1 Mark]
- A. Bva
 - B. $2Bva$
 - C. Bv/a
 - D. 0
- Q12. The induced electric field is: [1 Mark]
- A. Conservative
 - B. Non-conservative
 - C. Both conservative and non-conservative
 - D. Neither conservative nor non-conservative
- Q13. The SI unit of magnetic flux is: [1 Mark]
- A. Tesla
 - B. Weber
 - C. Henry
 - D. Ampere-meter

Section B - Short Answer Questions

Answer in 2-3 sentences. Each question carries 2 marks.

- Q14. State Faraday's laws of electromagnetic induction. [2 Marks]
- Q15. Explain Lenz's Law and show how it is a consequence of the law of conservation of energy. [2 Marks]
- Q16. Define self-inductance and state its S.I. unit. [2 Marks]
- Q17. Define mutual inductance and state its S.I. unit. [2 Marks]
- Q18. Derive the expression for motional EMF induced in a straight conductor of length ' l ' moving with velocity ' v ' in a uniform magnetic field ' B ' perpendicular to both ' l ' and ' v '. [2 Marks]
- Q19. Why is the self-inductance of a coil called 'electrical inertia'? [2 Marks]
- Q20. A metallic rod of length ' l ' is rotated with an angular frequency ' Ω ' about an axis passing through one of its ends and perpendicular to its length. A uniform magnetic field ' B ' (parallel to the axis) exists in the region. Find the EMF induced between the ends of the rod. [2 Marks]
- Q21. Distinguish between self-induction and mutual induction. [2 Marks]
- Q22. Explain the origin of eddy currents and mention one practical application. [2 Marks]
- Q23. A square loop of side 10 cm and resistance $0.5\ \Omega$ is placed in a magnetic field of 0.1 T perpendicular to its plane. If the field is reduced to zero in 0.1 s at a steady rate, find the magnitude of the induced current. [2 Marks]

Section C - Medium Answer Questions

Answer in 3-5 sentences. Each question carries 3 marks.

- Q24. (a) State Faraday's laws of electromagnetic induction. [3 Marks]
(b) A coil of N turns and area A is rotated at a constant angular velocity ' Ω ' in a uniform magnetic field B . Derive the expression for the induced EMF. (2 marks)
- Q25. (a) State Lenz's Law. [3 Marks]
(b) A conducting loop is held above a current-carrying long straight wire. If the current in the wire is suddenly increased, what will be the direction of the induced current in the loop? Justify your answer. (2 marks)
- Q26. Derive the expression for the self-inductance of a long solenoid of length L , area of cross-section A , and N turns. (3 marks) [3 Marks]

Q27. Derive the expression for the mutual inductance of two coaxial solenoids. (3 marks) [3 Marks]

Q28. A rectangular loop of sides 8 cm x 2 cm with a small cut is moving out of a region of uniform magnetic field of magnitude 0.3 T directed normal to the loop. What is the EMF induced across the cut if the velocity of the loop is 1 cm/s in a direction normal to the longer side? For how long does the induced EMF last? (3 marks) [3 Marks]

Q29. (a) What are eddy currents? [3 Marks]
(b) Mention two applications and two disadvantages of eddy currents. (2 marks)

Q30. A circular coil of radius 10 cm, 500 turns, and resistance $2\ \Omega$ is placed with its plane perpendicular to the horizontal component of the Earth's magnetic field. It is rotated about its vertical diameter through 180 degrees in 0.25 s. Estimate the magnitudes of the average EMF and current induced in the coil. (Horizontal component of Earth's magnetic field at the place = 3.0×10^{-5} T). (3 marks) [3 Marks]

Section D - Long Answer Questions

Answer in detail. Each question carries 5 marks.

Q31. (a) What is motional EMF? [5 Marks]
(b) A metallic rod PQ of length 'l' is moving with a velocity 'v' in a uniform magnetic field 'B' perpendicular to the rod and velocity. Derive the expression for the induced EMF.
(c) Show that the work done by the external agent in moving the rod is converted into electrical energy. (2 marks)

Q32. (a) Define self-inductance and mutual inductance. [5 Marks]
(b) Derive the expression for the energy stored in an inductor carrying current I.
(c) Two coils, P and Q, are placed close to each other. If the current in coil P is changed, will an EMF be induced in coil Q? Justify. (1 mark)

Q33. (a) State Faraday's laws of electromagnetic induction. [5 Marks]
(b) A square loop of side 'a' is placed in a uniform magnetic field 'B' perpendicular to its plane. The loop is then pulled out of the field with a constant velocity 'v'. Derive the expression for the induced EMF and the force required to pull it out.
(c) Plot a graph showing the variation of induced EMF as a function of time as the loop is pulled out. (1 mark)

Q34. (a) What is the phenomenon of mutual induction? [5 Marks]
(b) A pair of adjacent coils has a mutual inductance of 1.5 H. If the current in one coil changes from 0 to 20 A in 0.5 s, what is the change of magnetic flux linkage with the other coil?
(c) Explain why the self-inductance of a coil is always positive, while mutual inductance can be positive or negative depending on the relative orientation of the coils. (2 marks)